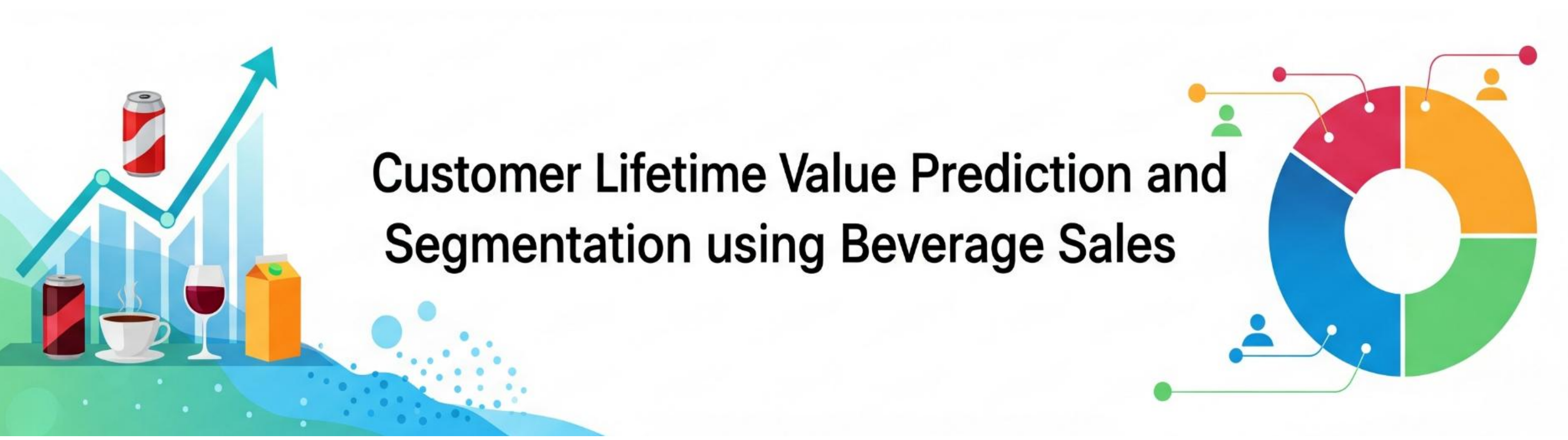

Project Overview

Customer Lifetime Value (LTV) prediction is a critical component in customer analytics. It helps businesses identify their most valuable customers, allocate resources effectively, and optimize marketing spend. This project focuses on building a machine learning–based LTV prediction pipeline, followed by customer segmentation to support data-driven decision-making.

Abstract

The project develops a predictive model to estimate Customer Lifetime Value using behavioral and transactional features such as recency, frequency, tenure, and spend. An XGBoost regression model was trained and achieved strong performance (**R^2: 0.986, RMSE: 1553.42**). Customers were segmented into four tiers (**Low, Lower-Mid, Upper-Mid, High**), providing actionable insights into revenue concentration and customer distribution. A Streamlit app was also built for interactive visualization and deployment.

Tools Used

- **Python Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, Joblib
- **Visualization:** Matplotlib, Seaborn
- **Interface:** Streamlit
- **Environment:** Jupyter Notebook, Python 3.13

Steps Involved

1. **Data Preprocessing**
   - Cleaned and transformed raw customer dataset.
   - Handled missing values and standardized variables.
2. **Feature Engineering**
   - Created log-transformed variables (Recency_log, Tenure_log, Frequency_log, AOV_log).
   - Derived Average Order Value (AOV) and other predictive features.
3. **Model Training**
   - Implemented XGBoost regression.
   - Evaluated model with multiple metrics (R^2, RMSE, MAE, MAPE).
4. **Segmentation**
   - Applied quartile-based segmentation on predicted LTV.
   - Created four groups: Low, Lower-Mid, Upper-Mid, High.
5. **Streamlit Deployment**
   - Built an interactive dashboard to visualize predictions, segment analysis, and business impact.

Results

- **Model Accuracy:**
  - **R^2** = 0.986
    The model explains **98.6% of the variance** in customer lifetime value (LTV). This is very strong, showing the features are highly predictive.
  - **RMSE** = 1553.42
    On average, predictions are off by about 1.5K units of LTV (in the same units as your target).
  - **MAE** = 900.34
    Typical prediction errors are about 900 units, which is quite low compared to average LTVs.
  - **MAPE** = 6.94%
    Predictions are within ±7% of actual values, which is considered very accurate in forecasting tasks.
- **Segmentation Insights:**
  - **25%** of customers classified as **"High"** segment contribute **~62%** of total revenue**.**
  - The **"Low"** segment contributes only **~6%.**
  - Differentiation highlights the importance of focusing retention and marketing on high-value customers.

Conclusion

The project successfully built a robust LTV prediction pipeline with high accuracy. By combining predictive modeling and segmentation, it provides business-ready insights to optimize customer strategy. The Streamlit app enables stakeholders to interact with the model outputs and make informed decisions.